📘 Overview of JavaScript

What is JavaScript?

JavaScript (JS) is a **high-level, interpreted, dynamic, client-side scripting language** primarily used to make web pages interactive.

It works together with:

- **HTML** → Structure
- **CSS** → Styling
- **JavaScript** → Behavior / Logic

Without JavaScript, web pages would be static (only text and images).

History of JavaScript

- Created in **1995**
- Developed by **Brendan Eich**
- Developed at **Netscape**
- Originally named **Mocha → LiveScript → JavaScript**
- Standardized as **ECMAScript**

⚠ JavaScript is NOT the same as Java.

Why JavaScript is Important

JavaScript is:

- ✅ The language of the web
- ✅ Supported by all browsers
- ✅ Used in frontend and backend
- ✅ Used in mobile & desktop apps
- ✅ Used in game development

Modern frameworks and libraries:

- **React**
- **Angular**
- **Vue.js**
- **Node.js**

Where JavaScript Runs

1. Browser (Client-Side)

Runs inside:

- **Google Chrome**
- **Mozilla Firefox**
- **Microsoft Edge**

Used for:

- Form validation
- DOM manipulation
- Event handling
- Animations

2Server (Backend)

Using:

- **Node.js**

Used for:

- API development
- Database operations
- Authentication systems

JavaScript Characteristics

🔹 **1. Interpreted Language**

Code runs line by line (no compilation required).

🔹 **2. Dynamic Typing**

You don't declare data types explicitly.

```
let x = 10;     // number
x = "Hello";    // now string
```

🔹 **3. Event-Driven**

Responds to user actions (click, hover, submit).

🔹 **4. Object-Oriented**

Supports objects, classes, prototypes.

🔹 **5. Asynchronous**

Can perform background tasks (API calls, timers).

JavaScript in Web Architecture

User → Browser → HTML + CSS + JavaScript → Server (Optional)

JavaScript interacts with:

- DOM
- APIs
- Browser Storage
- Server

How JavaScript Works in Browser

JavaScript Engine

Every browser has a JS Engine.

Example:

- **V8** (used in Chrome & Node.js)

Execution Steps:

1. Parsing
2. Compilation (Just-In-Time)
3. Execution

Basic Syntax Example

Example 1: Hello World

```
<script>
   console.log("Hello JavaScript");
</script>
```

Example 2: Button Click

```
<button onclick="showMessage()">Click Me</button>


<script>
function showMessage() {
   alert("Welcome to JavaScript!");
}
</script>
```

✓ This shows JavaScript makes web pages interactive.

Core Features of JavaScript

Feature	Description
Variables	Store data
Functions	Reusable code blocks
Objects	Data + behavior
Arrays	Store multiple values
DOM	Manipulate HTML
Events	User interactions
Async	Promises, async/await

JavaScript Versions

Important updates:

- ES5 (2009)
- ES6 (2015) – Major update
    - let, const
    - arrow functions
    - classes
    - modules
    - promises

Real-World Use Cases

JavaScript is used in:

- 🌐 Web Applications (Gmail-like apps)
- 📱 Mobile Apps (React Native)
- 🎮 Browser Games
- 🛒 E-commerce websites
- 📊 Dashboards & Data Visualization

Advantages of JavaScript

✓ Easy to learn
✓ Large community
✓ Runs everywhere
✓ Huge ecosystem
✓ Fast performance (V8 engine)

Limitations of JavaScript

❌ Browser security restrictions
❌ Can be disabled in browser
❌ Type-related bugs (due to dynamic typing)